

GRADE 9 PHYSICS · MOTION

Motion: Distance, Displacement and the Velocity-Time Graph

A build-it-from-the-ground-up book for two subtopics of Motion.

BAND B30 · building the foundation

GRADE 9

HINTS PRE-LOADED

WHO THIS BOOK IS FOR

A student who has met these words before but cannot yet draw the situation, cannot say what each symbol means without the formula, and is not sure whether an answer is sensible. Placement probes P1 and P2 were not passed. That is the normal starting point and this book is built for it: every worked example is complete, the rule is named only after you have already produced it, and support is given before it is asked for.

CONVENTIONS FIXED FOR THIS BOOK

Frame	Motion to the right is positive. The origin is where the object was when the stopwatch started.
Sign convention	right positive · left negative · time runs forward only
Symbols	d = distance — total path travelled; s = displacement — straight-line change in position, with direction; v = velocity at one instant; t = time
Source ledger	SRC-MOTION-v1
Band placement	Probes P1 (draw it) and P2 (say it) not passed → B30. Re-place after any prerequisite repair.

WHAT IS INSIDE

Subtopic	Title	Concept ID
M-01	Distance and Displacement	PHY-MOT-C01
M-02	Reading a Velocity-Time Graph	PHY-MOT-C03

HOW THIS BAND WORKS

Band B30 inverts the usual teaching order. You will see a full worked solution before you are asked to notice anything, because noticing a pattern requires something to notice it with. The first move is given to you and you are asked to justify it — that is a real thinking task and it is the right one at this stage.

HOW TO USE IT

Do this	Because
Run the prerequisite check before anything else	Both subtopics sit on a maths skill that is invisible until it fails. Two minutes here saves the whole subtopic.
Read the worked example without skipping lines	The line that looks obvious is usually the one that was missing.
Do the fades in order, do not jump to Fade 3	Each fade removes exactly one support. Skipping puts you back where you started.
Attempt the exit check with no numbers at all	If you can predict the direction without calculating, you own the idea.

M-01 · PHY-MOT-C01

Distance and Displacement

BAND B30 SRC §3.1-3.2

PREREQUISITE CHECK — RUN THIS FIRST

Ask: On a number line, you walk from the 2 mark to the 9 mark, then back to the 5 mark. How many marks did you cross altogether, and where do you end up?
If they cannot: Stop. The gap is directed number arithmetic on a line, not physics. Route to PHY-PREQ-NUMLINE-01 and return here afterwards. Teaching displacement on top of an open number-line gap produces the appearance of teaching and none of the effect.

source: SRC-MOTION-v1/§3.1

THE SITUATION

You leave your front door, walk 40 m up the street to the postbox, and then walk 15 m back towards home before stopping to talk to a friend.
Two different questions can be asked about that walk, and they have two different answers. How far did your feet travel? And how far are you now from your front door? This book is about why those are not the same question.

source: SRC-MOTION-v1/§3.1

PLACEHOLDER FIG-M01-01 status=RESERVED

- Horizontal street line, house at left marked 0 m, postbox at right marked 40 m
- Arrow 1: house → postbox, labelled 40 m, drawn above the line
- Arrow 2: postbox → 25 m mark, labelled 15 m, drawn below the line, arrowhead pointing left
- Final position marked with a standing figure at 25 m
- Dashed bracket from house to final position labelled 'displacement = 25 m'
- No numbers beyond those listed; no decorative scenery

reserve 40mm · src SRC-MOTION-v1/§3.1

The same walk, showing both the path taken and the straight-line change in position.

source: SRC-MOTION-v1/§3.1

WORKED — NOTHING LEFT OUT

The walk, worked completely

H3 GIVEN UP FRONT

System	You. Treated as a single point moving along a straight street. Nothing else in the picture matters.
Assumptions	The street is straight and you walk along it only. IDEALIZED_MODEL — good enough here because the street really is straight.
Frame and sign, in words	Away from home is the positive direction. Towards home is negative. The origin is your front door.
Diagram	See FIG-M01-01. Draw it yourself before reading on — this is the step that separates students who can do this from students who cannot.
Knowns	First leg: 40 m in the positive direction. Second leg: 15 m in the negative direction.
Distance — count the path	Distance adds up every metre your feet covered, and it never cares about direction. $40\text{ m} + 15\text{ m} = 55\text{ m}$.
Displacement — count the change in position	Displacement asks only where you started and where you finished. Start at 0. First leg takes you to +40. Second leg takes you back 15, so $40 - 15 = +25$.
Units check	Both answers came out in metres. Both are lengths, so that is right.
Sanity check	Displacement (25 m) must never be larger than distance (55 m). Your feet cannot cover less ground than the straight-line gap they opened up. If your displacement is ever bigger than your distance, you have made a mistake.

In one sentence

You walked 55 m in total, and you finished up 25 m from your door on the postbox side.

distance $d = 55\text{ m}$
displacement $s = +25\text{ m}$ (away from home)

source: SRC-MOTION-v1/§3.1, SRC-MOTION-v1/§3.2

NOW IT GETS ITS NAME

Distance is how much path you covered. Displacement is how much your position changed, and it carries a direction.
The names arrive now, attached to two numbers you have already worked out. That is deliberate: a name given before the thing it names is just a word to memorise.

source: SRC-MOTION-v1/§3.2

FADE — SUPPORT COMES OFF ONE STEP AT A TIME

Same idea, less help each time

Stage	What you are given	What you do
Fade 1	First line written for you: 'Distance = 30 + 30'	A cyclist rides 30 m east, then 30 m back west. Finish the distance, then find the displacement.
Fade 2	A drawn number line only	A dog runs 12 m right, 5 m left, then 4 m right. Find distance and displacement.
Fade 3	Nothing	A boy walks 60 m north, then 85 m south. Find distance and displacement, and say which direction he ends up from where he began.

working space

source: SRC-MOTION-v1/§3.2

NOTICE — YOUR FIRST REAL CHOICE

In Fade 3 the two numbers were 60 and 85. What in the question told you that the answer had to end up on the south side of where he started?
Looking for: The southward leg was longer than the northward one, so the negative direction won. The signs do the work, not the arithmetic.

source: SRC-MOTION-v1/§3.2

COMMON TRAP · PHY-M-021

The wrong idea: Displacement is just distance without the units, or distance rounded off — the two are basically the same number.
Why it is tempting: In every early example a student meets, the object only goes one way. When motion never reverses, distance and displacement really are equal, so the difference looks like fussy vocabulary rather than physics.
Probe: Give me a journey where the distance is 100 m and the displacement is zero.
Repair: Walk 50 m out and 50 m back. Your feet covered 100 m. Your position changed by nothing. Distance and displacement agree only when the motion never turns round — and turning round is exactly the case the exam will ask about.

source: SRC-MOTION-v1/§3.2

PRACTICE

On your own — same shape, new numbers

NO HELP GIVEN		
ID	Question	Type
P1	A cat walks 18 m east, then 7 m west. Find the distance and the displacement.	NEAR_TWIN
P2	A lift goes up 12 m, down 5 m, then up 2 m. Find the distance and the displacement.	NEAR_TWIN
P3	A boy walks 25 m south then 25 m north. Find the distance and the displacement.	NEAR_TWIN
P4	Write down a journey of your own where the distance is 60 m and the displacement is 20 m.	CONCEPT_REINFORCEMENT

working space

source: SRC-MOTION-v1/§3.2

EXIT CHECK — NO NUMBERS

Back to your walk to the postbox. Without any numbers: if you had walked all the way to the postbox and then all the way back to your front door, what would your distance be compared with before, and what would your displacement be? Say why.

your reasoning

source: SRC-MOTION-v1/§3.1

IF YOU GOT IT WRONG

If you got it wrong

What you wrote	What it means	Go to
Added 40 + 15 for displacement	CONCEPTUAL_GAP — treating displacement as a total	Re-read the name card, redo Fade 1
Got 25 but could not say which side of home	FRAME_SIGN_GAP	Sign drill PHY-DRILL-SIGN-01
Could not place the second leg on the line at all	PREREQUISITE_GAP	PHY-PREQ-NUMLINE-01 — stop this subtopic
Displacement came out bigger than distance	EXECUTION_GAP	Redo the sanity check step only

source: SRC-MOTION-v1/§3.2

YOU CAN MOVE ON WHEN YOU CAN DO ALL FOUR

1. Draw any straight-line journey on a number line and mark the start and finish. 2. Give the distance and the displacement for it, with a direction on the displacement. 3. Say in one sentence why the two are different. 4. Invent a journey where the displacement is zero and the distance is not. If any of these is shaky, redo Fade 2 and Fade 3 before starting M-02 — M-02 is built directly on displacement.

source: SRC-MOTION-v1/§3.2

M-02 · PHY-MOT-C03

Reading a Velocity–Time Graph

BAND B30 SRC §3.4

PREREQUISITE CHECK — RUN THIS FIRST

Ask: One square on this grid is 1 metre per second tall and 1 second wide. What do you get if you multiply those two together?

If they cannot: Stop. The answer is 1 metre, and if that is not obvious the gap is unit multiplication, not graphs. Route to PHY-PREQ-UNITALG-01. Everything below will otherwise be memorised as a ritual with no meaning attached.

source: SRC-MOTION-v1/§3.4

THE SITUATION

The school bus pulls away from the stop and drives at a steady 10 metres every second for 6 seconds before it reaches the junction.

You already know one way to find how far it went. This subtopic shows you a second way that looks completely different and gives exactly the same answer — and the second way keeps working when the first one breaks down.

source: SRC-MOTION-v1/§3.4

PLACEHOLDER FIG-M02-01 status=RESERVED

- Velocity–time axes. Vertical axis 'velocity (m/s)' 0 to 12, gridlines every 1
- Horizontal axis 'time (s)' 0 to 7, gridlines every 1
- Grid must be visible and countable — this figure exists to be counted, not admired
- Horizontal solid line at $v = 10$ from $t = 0$ to $t = 6$
- Region under the line lightly shaded
- One single square in the corner of the shaded region outlined in bold and labelled ' $1 \text{ m/s} \times 1 \text{ s} = 1 \text{ m}$ '
- No other annotation

reserve 52mm · src SRC-MOTION-v1/§3.4

The bus at a steady 10 m/s for 6 s. The grid is the point of this figure.

source: SRC-MOTION-v1/§3.4

WORKED — NOTHING LEFT OUT

How far did the bus go? Counted, then checked

H3 GIVEN UP FRONT

System	The bus, as a single point moving along a straight road.
Assumptions	The speed is exactly steady for the whole 6 s. Real buses are not this tidy; we choose a simple case so the idea is visible. IDEALIZED_MODEL.
Frame and sign, in words	Forward along the road is positive. The bus never reverses here, so everything stays positive.
Diagram	FIG-M02-01. Velocity up the side, time along the bottom, grid drawn so one square is 1 m/s tall and 1 s wide.
Knowns	$v = 10 \text{ m/s}$ and it does not change. t runs from 0 to 6 s.
Now count	Count the squares underneath the line. There are 10 rows of them and 6 columns. That is 60 squares.
What is one square worth?	$1 \text{ m/s across} \times 1 \text{ s along} = 1 \text{ metre}$. That is the check you did at the top of this page. So 60 squares means 60 metres.
Check it against what you already know	$10 \text{ m/s for } 6 \text{ s is } 10 \times 6 = 60 \text{ m}$ — speed times time, the rule you already had. The counting and the old rule agree, because they are the same statement written two ways.
Units check	$\text{m/s} \times \text{s} = \text{m}$. The answer is a length. Correct.

Sanity check	60 m is about three bus lengths and a bit — believable for six seconds of town driving. If you had got 600 m or 6 m, something went wrong.
In one sentence	In those six seconds the bus covered sixty metres of road.

counted area = 60 squares = 60 m
speed × time = 10 × 6 = 60 m

source: SRC-MOTION-v1/§3.4

NOW IT GETS ITS NAME

The area underneath a velocity–time graph gives the displacement.
You have now produced 60 m twice, by two routes, before this sentence appeared. The sentence is a label for something you already own — not a new fact to memorise.

source: SRC-MOTION-v1/§3.4

FADE — SUPPORT COMES OFF ONE STEP AT A TIME

Same idea, less help each time

Stage	What you are given	What you do
Fade 1	Instruction given: 'count the squares under the line'	A scooter travels at a steady 8 m/s for 5 s. Sketch the graph and find the distance.
Fade 2	The graph is drawn for you, nothing else	A runner holds 4 m/s for 7 s. Find the distance.
Fade 3	Nothing	A cyclist rides at a steady 6 m/s for 9 s. Draw the graph and find the distance.
Fade 4 — new shape, first move given	First move given: 'the region under the line is a triangle, and you already know how to find a triangle's area'	A car starts from rest and speeds up steadily to 12 m/s over 4 s. Find how far it travelled.

working space

source: SRC-MOTION-v1/§3.4

NOTICE — YOUR FIRST REAL CHOICE

In Fade 4 you could not just multiply the two numbers on the axes the way you did in Fade 1. What was different about that graph that told you so?
Looking for: The line sloped, so the velocity was not one single number for the whole journey — there was no single speed available to multiply by.

source: SRC-MOTION-v1/§3.4

COMMON TRAP · PHY-M-008

The wrong idea: The height of the graph tells you how far the object has gone.

Why it is tempting: On a distance–time graph the height genuinely is the distance, and that graph is usually taught first. The eye learns to read height as 'how far' and carries the habit onto a graph where the vertical axis means something else entirely.

Probe: Point at the line at $t = 3\text{ s}$ on FIG-M02-01 and tell me what the number 10 there is measuring.

Repair: On this graph the height is the speed at that instant, not the distance. The distance is stored in the area underneath. Check the axis label before you read any value off a graph — the label tells you which of the two you are looking at.

source: SRC-MOTION-v1/§3.4

PRACTICE

On your own — same shape, new numbers

NO HELP GIVEN		
ID	Question	Type
P5	A tram holds 9 m/s for 5 s. Sketch the graph and find the distance.	NEAR_TWIN
P6	A skateboarder holds 3 m/s for 8 s. Sketch the graph and find the distance.	NEAR_TWIN
P7	A train starts from rest and speeds up steadily to 20 m/s over 10 s. Find the distance.	NEAR_TWIN
P8	Two graphs both cover 40 m. One is a rectangle, one is a triangle. Sketch a possible pair.	CONCEPT_REINFORCEMENT

working space

source: SRC-MOTION-v1/§3.4

EXIT CHECK — NO NUMBERS

Back to the bus. Suppose it pulls away and reaches the same junction at the same moment as before, but this time it speeds up gradually from rest instead of holding 10 m/s the whole way. Did it travel further, less far, or exactly the same distance? Say how you know — no numbers, no calculating.

your reasoning

source: SRC-MOTION-v1/§3.4

IF YOU GOT IT WRONG

If you got it wrong

What you wrote	What it means	Go to
Answered 16 (added 10 and 6)	EXECUTION_GAP	Redo Fade 1 with the first line given
Counted the squares but could not say why each is a metre	PREREQUISITE_GAP — unit algebra	PHY-PREQ-UNITALG-01 — stop this subtopic

What you wrote	What it means	Go to
Read the height (10) as the distance	REPRESENTATION_GAP	Trap PHY-M-008 above, then Fade 2
Used $\frac{1}{2} \times \text{base} \times \text{height}$ on the rectangle	MODEL_SELECTION_GAP	Shape-recognition pair PHY-PAIR-SHAPE-01
Answered '60 metres per second'	EXECUTION_GAP — units	Dimensional check drill PHY-DRILL-DIM-01

source: SRC-MOTION-v1/§3.4

SOMETHING YOU ARE BEING TOLD HONESTLY

The 'steady speed' in these examples is a simplification. Real buses wobble a metre per second either way, and later you will meet graphs that curve instead of running flat or straight. Nothing you learned here will need to be thrown away — the area still gives the displacement. It just becomes an area with a curved edge.

YOU CAN MOVE ON WHEN YOU CAN DO ALL FOUR

1. Sketch a velocity-time graph for a steady speed and find the distance from it. 2. Say what one square on the grid is worth, and why. 3. Do the same for a graph that ramps up from rest. 4. Explain, without numbers, why reading the height off a velocity-time graph does not give you the distance. If number 4 is shaky, that is the trap PHY-M-008 above and it is the one most likely to cost you marks.

source: SRC-MOTION-v1/§3.4

APPENDIX

Source, link and badge ledger

Every teaching block traces to a source obligation or is declared VALUE_ADD. Nothing in this book is unattributed.

Subtopic	Block	Source obligation
M-01	prereq_check	SRC-MOTION-v1/§3.1
M-01	purpose_anchor	SRC-MOTION-v1/§3.1
M-01	figure	SRC-MOTION-v1/§3.1
M-01	worked	SRC-MOTION-v1/§3.1, SRC-MOTION-v1/§3.2
M-01	name_card	SRC-MOTION-v1/§3.2
M-01	fade_ladder	SRC-MOTION-v1/§3.2
M-01	notice	SRC-MOTION-v1/§3.2
M-01	misconception	SRC-MOTION-v1/§3.2
M-01	practice	SRC-MOTION-v1/§3.2
M-01	exit_check	SRC-MOTION-v1/§3.1
M-01	gap_map	SRC-MOTION-v1/§3.2
M-01	callout	SRC-MOTION-v1/§3.2
M-02	prereq_check	SRC-MOTION-v1/§3.4
M-02	purpose_anchor	SRC-MOTION-v1/§3.4
M-02	figure	SRC-MOTION-v1/§3.4
M-02	worked	SRC-MOTION-v1/§3.4
M-02	name_card	SRC-MOTION-v1/§3.4
M-02	fade_ladder	SRC-MOTION-v1/§3.4
M-02	notice	SRC-MOTION-v1/§3.4
M-02	misconception	SRC-MOTION-v1/§3.4
M-02	practice	SRC-MOTION-v1/§3.4
M-02	exit_check	SRC-MOTION-v1/§3.4
M-02	gap_map	SRC-MOTION-v1/§3.4
M-02	callout	VALUE_ADD
M-02	callout	SRC-MOTION-v1/§3.4

Figure placeholders awaiting artwork

Figure	Status	Reserved	Caption
FIG-M01-01	RESERVED	40 mm	The same walk, showing both the path taken and the straight-line chang
FIG-M02-01	RESERVED	52 mm	The bus at a steady 10 m/s for 6 s. The grid is the point of this figu

Audit checkpoints cleared at render time

Checkpoint	Result
BAND DECLARED (SRU-23)	PASS — B30 on every block
BAND FIDELITY D1	PASS — worked-before-notice order held in both subtopics
FRAME/SIGN BEFORE ALGEBRA (SRU-24)	PASS — declared on cover and restated in every worked block
SYMBOL DRIFT D2	PASS — d, s, v, t used per frozen table throughout
VOCABULARY DRIFT D8	PASS — distance/displacement and speed/velocity kept strict
PREREQUISITE ROUTED (SRU-26)	PASS — 2 checks, both name a destination concept
GAP MAP COVERAGE (SRU-19)	PASS — every subtopic ends in a gap map
MODEL UPGRADE RECORDED (SRU-25)	PASS — idealisation flagged in M-02 callout
SOURCE LINKAGE	PASS — 21 of 22 blocks cite an obligation; 1 declared VALUE_ADD
PREVIEW LEAKAGE	N/A — no GRADE_11_PREVIEW at B30, as required
FIGURE PLACEHOLDERS	2 reserved, bounds fixed, semantic spec complete
BAND PRACTICE MIX	PASS — 4 near-twin + 1 reinforcement per subtopic, per B30 contract